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Muscle synergies involved in preparation to a step made under the self-paced and reaction time instructions

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Abstract

Objective: To investigate multi-muscle synergies during preparation to making a step and their changes with speed of stepping and method of step initiation, self-paced vs. reaction-time.

Methods: Standing healthy persons performed stepping tasks under 3 instructions, 'comfortably, self-paced', 'very quick, self-paced', and 'as fast as possible to a visual signal'. Electromyographic (EMG) signals of 10 postural muscles were recorded and analyzed. The participants also performed standard tasks associated with releasing a load. These trials were used to identify muscle groupings (M-modes) associated with shifts of the center of pressure (COP) and relations between small changes in the M-modes and COP shifts. The framework of the uncontrolled manifold hypothesis was used to partition the EMG variance across stepping trials into two components that kept constant and changed the COP coordinates in the anterior–posterior (AP) and medio-lateral (ML) directions.

Results: Muscle synergies stabilizing COP trajectories in both AP and ML directions were observed in both the stepping and supporting legs. The synergies tended to weaken and disappear close to the moment of take-off. They showed only minor differences across the 3 tasks. In contrast, mechanical and EMG characteristics of postural adjustments to stepping were significantly larger and showed higher rates of change during quick steps and steps under the reaction-time instruction.

Conclusions: COP shifts associated with preparation to making a step are stabilized by multi-muscle synergies. These synergies are robust across stepping speed and methods of step initiation, self-paced and reaction-time.

Significance: Studies of multi-muscle postural synergies are important for understanding postural disorders in neurological patients and changes in postural strategies with age.

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1. Introduction

When a standing person makes a step, there are early changes in the activity of postural muscles and associated shifts of the center of pressure (COP, the point of application of the resultant force acting on the body from the supporting surface) in both anterior–posterior (AP) and medio-lateral (ML) directions (Elble et al., 1994). These changes can be seen up to 1 s prior to the take-off of the stepping foot. Their functional role has been assumed to unload the stepping leg and to produce a moment of the reaction force rotating the body forward (Ito et al., 2003; Koryakin, 1958—cited after Zatsiorsky, 1966; Lepers and Breniere 1995).

Studies of muscle synergies involved in preparation to making a step may be of substantial clinical importance. They may able to provide important information related to the early development of stepping in toddlers and decline of the locomotor function in elderly or in patients with postural disorders (Assaiante et al., 2000; DiFabio et al., 2005; Melzer and Oddson 2004; Rogers et al., 2001). In particular, postural preparation processes have been shown to be impaired in patients with Parkinson's disease (Bazalgette

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et al., 1986; Diener et al., 1989); however, these processes have not been quantified at the level of multi-muscle synergies.

Recently, a particular quantitative method has been developed to analyze multi-muscle synergies based on changes in the activation patterns (electromyographic signals, EMGs). This method is based on the uncontrolled manifold (UCM) hypothesis (Scholz and Schoner, 1999; reviewed in Latash et al., 2002), which assumes that the neural controller acts in a space of independent elemental variables and creates in that space a sub-space (UCM) corresponding to a desired value of an important performance variable. Under independent elemental variables, we imply a set of variables at a selected level of analysis that are assumed to be manipulated by the central nervous system to stabilize performance variables that may be task or intention specific. Then, variability of elemental variables is structured to be mostly constrained to the UCM thus stabilizing the performance variable. The method was applied to analysis of postural synergies stabilizing COP shifts with elemental variables associated with muscle groups (M-modes) scaled together over a group of tasks (Krishnamoorthy et al., 2003a,b, 2004).

In an earlier study, we applied this method to analysis of COP shifts prior to step initiation and found that COP trajectories in both AP and ML directions were stabilized across trials by co-varied changes in M-modes (Wang et al., in press). In the current study we address two questions. First, are multi-muscle synergies robust with respect to preparation to steps performed at a natural pace and very quickly, both in a self-paced manner? Second, are there changes in the multi-muscle synergies when a step is performed under time pressure, i.e. under a typical simple reaction time instruction?

With respect to the first question, it has been shown that mechanical and EMG patterns of postural adjustments prior to a quick step differ from those seen prior to a comfortable step (Couillandre et al., 2002; Ito et al., 2003). A change in average characteristics of postural adjustments may or may not be associated with a change in the underlying synergies. With respect to the second question, anticipatory postural adjustments (APAs, reviewed in Massion, 1992) prior to a voluntary movement performed by a standing person show time shifts and changes in their pattern under the simple reaction time instruction (Boisset and Zattara, 1990; De Wolf et al., 1998; Lee et al., 1987). Assembling synergies has also been shown to take time of the order of a few hundred milliseconds, although in studies of multi-finger interaction (Shim et al., 2003). Hence, we would expect that when a person makes a step as quickly as possible following an imperative visual signal the multi-muscle synergies emerge later in time than during the self-paced movements.

We used two tasks in the study, a load release task associated with reproducible APAs (Aruin and Latash, 1996) and a stepping task. The load release task was used for identification of M-modes; it has a certain advantage as compared to voluntary movement tasks since the magnitude of the induced perturbation does not vary with the natural variability in the voluntary movement that triggers the load release. Using different tasks at the two stages of the analysis allows to avoid circular logic and provides additional support for the idea of stable M-modes used over a variety of tasks associated with COP shifts.

2. Methods

2.1. Subjects

Eight healthy subjects, 6 males and two females, mean weight 71.3 kg (±12.3 SD), mean height 176.3 cm (±10.5 SD) and mean age 33.0 years (±9.6 SD), without any known neurological or motor disorder, participated in the experiment. All subjects were right-handed according to their hand usage during eating and writing. All of them were right foot dominant. The tests used to determine the dominant foot included kicking a ball, stepping up on a chair, and leading off in the long jump. The subjects gave written informed consent according to the procedure approved by the Office for Regulatory Compliance of The Pennsylvania State University.

2.2. Apparatus

In all conditions, the subjects stood with socks on a force platform (AMTI, OR-6, Watertown, MA, USA), which was used to record the moment around the sagittal–horizontal axis (M_x), the moment around the frontal–horizontal axis (M_y), the vertical component of the reaction force (F_z), and the shear force (F_x) in the anterior–posterior direction. A foot switch was placed under the metatarsal area of each foot to measure the timing of toe off.

In some conditions the subjects held a light-weight handle in front of their body by pressing on its sides with both hands, while in other conditions, they held a handle by the right hand as illustrated in Fig. 1A. These handles were connected to a load via a pulley system such that the load created a force acting upwards (Fig. 1A). A uni-directional accelerometer (Sensotec, Columbus, OH, USA) was taped on the load and used for trials alignment (see later). The axis of sensitivity of the accelerometer was directed along the motion of the load. A 2-m long and 1-m wide walkway was leveled with the platform. An electric bulb used as an imperative stimulus in reaction time experiments (see below) was mounted 3 m away from the platform, at the subject's eye level.

Disposable self-adhesive electrodes (3M) were used to record the surface EMG of the following postural muscles from the right side of the body: lateral head of gastrocnemius (GL), soleus (SOL), biceps femoris (BF), gluteus medius (GMED), erector spinae (ES), tibialis anterior (TA), vastus lateralis (VL), rectus femoris (RF),

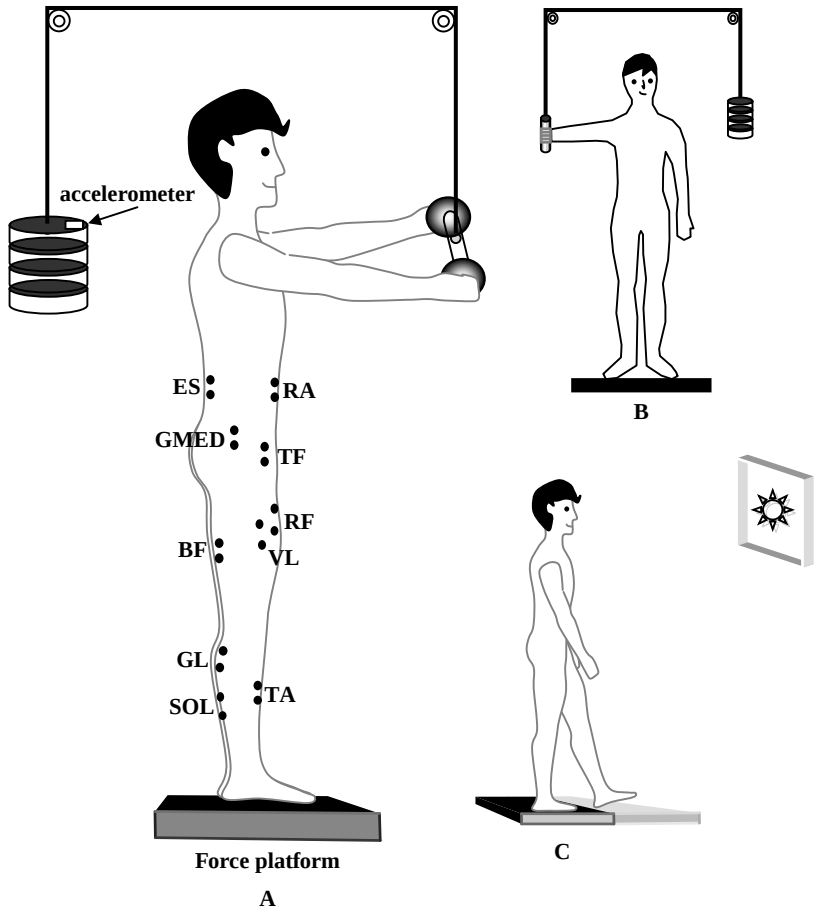

Fig. 1. The experimental setup: (A) Subjects stood on the force platform. In trials when subjects were required to release the load behind them (LR_B), they pressed with the hands on the round ends of a circular handle, which was connected to the load through a pulley system. (B) When subjects were required to release the load by their body side (LR_S), they held with the right hand a handle attached to the load through a pulley system. Location of some of the EMG electrodes is shown in A (GL, lateral head of gastrocnemius; GM, medial head of gastrocnemius; SOL, soleus; ST, semi-tendinosus; BF, biceps femoris; GMED, gluteus medius; ES, erector spinae; TA, tibialis anterior; VL, vastus lateralis; RF, rectus femoris; TF, tensor fasciae latae; RA, rectus abdominis). (C) In the stepping task (ST), the subjects were required to make a single step forward in a self-paced natural manner (ST_{CS}), very quickly (ST_{QS}), or under a typical simple reaction time instruction, i.e. as quickly as possible following a visual signal (ST_{RT}).

tensor fasciae latae (TF), and rectus abdominis (RA). The electrodes were placed on the muscle bellies, with their centers 3 cm apart. In addition, a reference electrode was attached to the lateral aspect of the fibula. The EMG signals were amplified by means of differential amplifiers ($\times 3000$). The data were digitized at the sampling frequency of 1000 Hz with a 12-bit resolution. A Gateway 450 MHz PC with customized software based on the LabView-5 package (National Instruments, Austin, TX, USA) was used to control the experiment and collect the data.

2.3. Procedures

Five types of tasks were used. Two tasks were associated with APAs and involved COP shifts as an implicit component. The first task required the subject to release the load (load release task, LR) suspended behind him/her using the pulley system, LR_B (cf. Aruin and Latash, 1996; Krishnamoorthy et al,. 2003a,b; see Fig. 1A). The second task required the subject to release the load suspended by his/her left side of the body, LR_S (Fig. 1B). These two tasks were used to explore EMG combinations responsible for anterio-posterior and lateral COP shifts, respectively.

The other 3 tasks required the subjects to make a single step forward. One of them required the subject to make a comfortable step from quiet stance in a self-paced manner (ST_{CS}). Another task required the subjects to make a self-paced very quick step, ST_{QS}. Within the final task, the subjects were to make a step under a typical simple reaction time instruction, i.e. as quickly as possible following a visual signal (ST_{RT}).

2.3.1. Load release in the back (LR_B)

In the initial position, the subject stood on the force platform with his/her feet side-by-side, at hip width. This position was marked on the platform and was reproduced across trials. The subject held a load suspended behind the body by pressing with the hands of the extended arms on the two round ends of the light horizontal handle. The upward acting load was used to avoid excessive fatigue associated

with holding the combined weight of the arms and the load. The load mass ranged from 2.2 to 5 kg (3–10% of subject's body mass, on average), in increments of 0.46 kg. Subjects were instructed to release the load in a self-paced manner with quick, small amplitude, bilateral shoulder abduction movement. The total of 63 trials was performed; those included 9 repetitions (presented as a block) of release of each of the 7 loads in a balanced order. Prior to each task, two practice trials were given.

2.3.2. Load release at the side (LR_S)

The initial position of the trunk and the legs was the same as in the LR_B task. The subject's left hand was hanging loosely by his/her left side. The subject grasped a wooden handle with the right hand; the right shoulder was adducted, and the right arm fully extended in a frontal plane (Fig. 1B). The handle was connected via a pulley system to a load suspended at the left side of the subject's body. The mass of the load ranged from 1.2 to 4 kg, in increments of 0.46 kg. The subject was asked to release the handle in a self-paced manner with a quick hand-opening motion of the right hand. Prior to each task, two practice trials were given. The total of 63 trials was performed; those included 9 repetitions (presented as a block) of release of each of the 7 loads in a balanced order.

2.3.3. Comfortable stepping task (ST_{CS})

The subjects began each trial by standing naturally and quietly on the force platform, the hands hanging loosely by the sides. The initial position of the feet was the same as in the LR tasks. In response to a beep cue, the subjects were free to initiate a forward step in a comfortable self-paced manner. After two practice trials, twenty data collection trials were performed. There were two series with steps performed by the right leg and by the left leg; the order was balanced across subjects.

2.3.4. Quick stepping task (ST_{QS})

The subjects began each trial by standing in the same position as in the ST_{CS} task. In response to a beep cue, the subjects were free to initiate a forward very quick step in a self-paced manner. After two practice trials, 40 trials were collected in two blocks with steps performed by the left leg and by the right leg.

2.3.5. Reaction time stepping task (ST_{RT})

The subjects began each trial by standing in the same position as in the other two stepping tasks. There was a beep generated by the computer, which served as a warning signal. The imperative light signal was turned on at unpredictable delays ranging from 1.5 to 4 s. The light was placed on the wall directly in front of the subjects, at the eye level, 3 m away. After 3–5 practice trials, 50 trials were collected in two blocks of 25 trials each with steps performed by the left leg and by the right leg in a balanced order.

For each trial, data were collected over 5 s. Subjects were instructed to stand as quietly as possible in the initial position before the beginning of the trial. The subjects heard a computer generated beep 500 ms after data collection had begun. In the SR_{CS} and ST_{QS} series, the subjects were reminded not to initiate their actions immediately after the beep, but to wait for about 1 s. In these series, the subjects were asked to look at a mark placed on the wall directly in front of them. The mark was placed at the eye level, 3 m away. The ambient light was dimmed during all stepping tasks.

The order of the ST tasks was always the same, from the easiest (ST_{CS}) to the hardest (ST_{RT}). This was done to assure optimal performance of the subjects across all tasks. Rest periods of 6 s between trials and 2 min between conditions were given. Additional rest periods (about 1 min) were given between sets of trials within a task, such that fatigue was never an issue.

In order to compare indices of muscle activation across subjects, 3 control trials were performed. In two of these trials, the subjects were asked to hold a load of 5.5 kg quietly for 5 s in front of the body and behind the body through the pulley system. In the third trial, the subjects held the 5.5 kg load quietly for 5 s by grasping a wooden handle with the right hand; the right shoulder was adducted, and the right arm fully extended in a frontal plane. The handle was connected via a pulley system to a load suspended at the left side of the subject's body. These data were used for EMG normalization as described in Section 2.4. Different maximal loads in different tasks were used to optimize the load range and simultaneously not to overstrain the subjects (based on pilot tests). In particular, all subjects could hold in two hands for a few seconds the 5.5 kg load. However, in repetitive trials, we limited the maximal load to 5.0 kg since repeating the task with the 5.5 kg load was hard for some of the subjects. During the load release trials at the body side, the subjects used only one hand to hold the load. Therefore, the maximal load value was further decreased (4.0 kg).

2.4. Data processing

All signals were processed off-line using LabView-5 software. All EMG signals were rectified and filtered with a 50 Hz low-pass, second order, zero-lag Butterworth filter. All kinetic signals were filtered with a 20 low-pass, second order, zero-lag Butterworth filter. Individual LR_B and LR_S trials were viewed on a monitor screen and aligned according to the first change in the signal from the accelerometer (time zero, t_0). When a voluntary movement started, the accelerometer showed an abrupt change in its signal, which was defined visually. ST trials were aligned by the toe-off time, which was defined using the signal from the foot switch.

Changes in the muscle activity associated with the early phase of the COP shift were quantified as follows. For the LR_B and LR_S trials, rectified EMG signals were integrated from 100 ms prior to t_0 up to t_0 ($\int$EMG). These values were

corrected by EMG integrals quantified over a 100 ms time interval 0.5 s prior to t_0 (see Eq. (1a); Krishnamoorthy et al., 2003b):

$$\mathrm{QS}: \Delta I\,\mathrm{EMG} = \int_{t_0-100}^{t_0} \mathrm{EMG}\,\mathrm{d}t - \int_{t_0-600}^{t_0-500} \mathrm{EMG}\,\mathrm{d}t \tag{1a}$$

$$\mathrm{ST}: \Delta I\mathrm{EMG}_j = \int_{t_0-100*j}^{t_0-100(j-1)} \mathrm{EMG}\,\mathrm{d}t - \int_{t_0-1100}^{t_0-1000} \mathrm{EMG}\,\mathrm{d}t \tag{1b}$$

For the ST trials, rectified EMG signals were integrated over 100 ms time intervals starting 800 ms prior to t_0 and ending up at t_0. These EMG integrals were corrected by subtracting the EMG integrals over a 100 ms time interval 1 s prior to t_0, Eq. (1b). The eight 100 ms time intervals are referred to in Eq. (1b) with the subscript $j=1,\ldots,8$. These intervals were selected as a compromise between avoiding to deal with poorly reproducible sample-to-sample changes in the EMG signals and getting insight into possible time development of the multi-muscle synergies.

In order to compare the ΔIEMG indices across muscles and subjects, we normalized them by the maximal EMG integrals over the control trials as follows: ΔIEMG indices for the dorsal (SOL, GL, BF, and ES), ventral (TA, RF, VL, and RA), and lateral (TFL and GMED) muscles were divided by the EMG integrals over 100 ms ($I\mathrm{EMG_C}$) in the middle of the corresponding control trial, that is during holding the load quietly in front of the body, behind the body, or at the body side, respectively:

$$\Delta I\mathrm{EMG_N} = \Delta I\mathrm{EMG}/I\mathrm{EMG_C} \tag{2}$$

Coordinates of the center of pressure (COP) in the anterio-posterior (AP) and medio-lateral (ML) directions can be computed as:

$$\mathrm{COP_{AP}} = (-M_Y - F_X d)/F_Z$$

$$\mathrm{COP_{ML}} = (M_X - F_Y d)/F_Z$$

where d is the vertical distance from the origin of the platform to its surface (36 mm according to the platform specifications). Our estimations of the contribution of changes in the shear forces to the COP shifts suggest that they were always under 8%. Therefore, COP shifts were calculated using the following approximations:

$$\Delta\mathrm{COP_{AP}} = -\Delta M_Y/F_Z \tag{3a}$$

$$\Delta\mathrm{COP_{ML}} = \Delta M_X/F_Z \tag{3b}$$

For the $\mathrm{LR_B}$ and $\mathrm{LR_S}$ trials, COP shift corresponding to the EMG integral indices defined above was computed as the difference between the COP position 50 ms following the interval of EMG integration and the average COP position measured within a 50 ms time interval prior to the period of EMG integration. This comparison assumes an average 50 ms electromechanical delay (cf. Krishnamoorthy et al., 2003a).

For the ST trials, COP shifts and F_X changes were also quantified within the time interval {−1000 ms; 0 ms} with respect to t_0. The difference between the COP coordinates at −1000 ms and at t_0 was computed for the AP and ML directions separately ($\Delta\mathrm{COP_{AP}}$ and $\Delta\mathrm{COP_{ML}}$). In addition, COP trajectory was differentiated with respect to time within the same time interval for the AP and ML directions separately. The peak rate of COP shift was defined. The onset of COP shift was defined as the time when the COP derivative first exceeded 5% of its peak value. F_X changes were quantified in a similar way. That is, the change in F_X within the time interval {−1000 ms; 0 ms} with respect to t_0, its peak derivative, and the time when F_X started to change were defined using the same criteria as described for COP shifts. The reaction time was defined as the time period from the onset of the visual cue to the toe-off time.

2.5. Statistics

Descriptive statistical methods were used. Data are presented as means and standard errors.

2.5.1. Step 1: Defining M-modes using principal component analysis (PCA)

For the $\mathrm{LR_B}$ and $\mathrm{LR_S}$ tasks, for each subject, we got $\Delta I\mathrm{EMG_N}$ data matrices with the size 63×10 (63 rows corresponding to the trials and 10 columns corresponding to the muscles). The correlation matrix among the $\Delta I\mathrm{EMG_N}$ was subjected to PCA, using procedures from SPSS 10.0 (SPSS, Inc., Chicago, IL, USA). The factor extraction module with principal component extraction was employed.

For each subject, the obtained eigen-values and PCs were then considered. The first 3 PCs for the $\mathrm{LR_B}$ and $\mathrm{LR_S}$ tasks, and the first 4 PCs for some of the $\mathrm{LR_S}$ tasks (described in more detail in Section 3) were selected for further analysis. This was based on two criteria, examination of the scree plots and having at least two muscle significantly loaded per PC (see Krishnamoorthy et al., 2003b). We are going to address these PCs as muscle modes (M-modes) and assume that magnitudes of (coefficients at) the M-modes are manipulated by the controller to produce COP shifts. Three M-modes were defined for the $\mathrm{LR_B}$ task, further addressed as $\mathrm{M_{1B}}$, $\mathrm{M_{2B}}$, and $\mathrm{M_{3B}}$. For the $\mathrm{LR_S}$ task, 3 M-modes were defined, further addressed as $\mathrm{M_{1S}}$, $\mathrm{M_{2S}}$, and $\mathrm{M_{3S}}$. In 6 subjects, a fourth M-mode ($\mathrm{M_{4S}}$) was accepted if it had at least two muscles significantly loaded (loadings over 0.5; Hair et al., 1995).

2.5.2. Step 2: Defining the Jacobian using multiple regression

Linear relations between changes in the magnitudes of M-modes (ΔM) and the COP shifts (ΔCOP) were assumed and the corresponding multiple regression equations computed over the trials performed by each subject

separately. The coefficients of each regression equation were arranged in a matrix that is a Jacobian matrix. The LR_B and LR_S tasks were used to generate linear estimates of the following Jacobians ($\mathbf{J}_{AP}$ and $\mathbf{J}_{ML}$)

$$\Delta COP_{AP} = k_1\Delta M_{1B} + k_2\Delta M_{2B} + k_3\Delta M_{3B}; \quad J_{AP} = [k_{1B}k_{2B}k_{3B}]^T, \tag{4A}$$

$$\Delta COP_{ML} = k_1\Delta M_{1S} + k_2\Delta M_{2S} + k_3\Delta M_{3S}; \quad J_{ML} = [k_{1S}k_{2S}k_{3S}]^T, \tag{4B}$$

$$\Delta COP_{ML} = k_1\Delta M_{1S} + k_2\Delta M_{2S} + k_3\Delta M_{3S} + k_4\Delta M_{4S}; \quad J_{ML} = [k_{1S}k_{2S}k_{3S}k_{4S}]^T, \tag{4C}$$

where subscripts B and S correspond to LR_B and LR_S tasks, respectively, while subscripts AP and ML correspond to COP shift in the anterio-posterior and medio-lateral directions, respectively. Within this approach, the **J** matrices are reduced to (3×1) and (4×1) vector-columns.

2.5.3. Step 3: UCM analysis

For each trial of the ST tasks, $\Delta IEMG_N$ were computed and transformed into ΔMs using the results of Step-1 of the analysis. The ΔM space has dimensionality $n_{AP}=3$ for the COP_{AP} shifts and $n_{ML}=3$ or 4 for the COP_{ML} shifts. A hypothesis that a particular magnitude of ΔCOP is stabilized by covariation of magnitudes of the M-modes accounts for one degree of freedom ($d=1$). Thus, the system is redundant with respect to the task of stabilizing particular ΔCOP values. The mean magnitudes of each ΔM for a selected time interval across a series of ST trials were computed. Since the model relating ΔMs to ΔCOP is linear, the mean values were subtracted from each computed value and the residuals were further analyzed as follows.

The UCM represents combinations of M-modes that are consistent with a stable value of ΔCOP. The UCM was calculated as the null space of the corresponding **J** matrix. The null space of **J** is a set of all vector solutions **x** of a system of equations $\mathbf{Jx}=0$. The null space is spanned by basis vectors, $\boldsymbol{\varepsilon}_i$. The vector of individual mean-free ΔMs was resolved into its projection onto the null space:

$$f_{UCM} = \sum_{i=1}^{n-d} \left(\underline{\varepsilon}_i^T(\Delta M)\right)\underline{\varepsilon}_i \tag{6A}$$

and component orthogonal to the null space:

$$f_{ORT} = (\Delta M) - f_{UCM} \tag{6B}$$

The amount of variance per DOF within the UCM is

$$V_{UCM} = \sigma^2_{UCM} = \sum_{i=1}^{N} f^2_{UCM}/\left((n-d)N_{trials}\right) \tag{7A}$$

and orthogonal to the UCM is:

$$V_{ORT} = \sigma^2_{ORT} = \sum_{i=1}^{N} f^2_{ORT}/(dN_{trials}) \tag{7B}$$

V_{UCM} and V_{ORT} were the main dependent variables used in the study. To quantify the relative amount of the total variance that is compatible with stabilization of a particular shift of the COP we used a variable (ΔV) reflecting the difference between the variance within the UCM and orthogonal to the UCM. It was computed as:

$$\Delta V = (V_{UCM} - V_{ORT})/V_{TOT} \tag{8}$$

where all variance indices are computed per degree of freedom; V_{TOT} means the total variance.

Friedman non-parametric test was used with factors *Condition* (ST_{CS}, ST_{QS} and ST_{RT}) and *Time interval* (8 intervals) to analyze changes in ΔV across tasks and with time. Different time intervals from −800 ms prior to t_0 up to t_0 allow getting insight into possible time development of the multi-muscle synergies and stability of the muscle modes. Analysis was performed separately for COP shifts in the anterior–posterior and medio-lateral directions. Level of significance was set at $P=0.05$.

3. Results

3.1. Identification of M-modes

3.1.1. General EMG patterns in the LR task

When a subject stood and held a load behind the body (LR_B task), there was increased activity of the ventral muscles (TA, VL, RF, TF, and RA). Prior to load release, the activity in these muscles typically dropped, and there could also be EMG bursts in dorsal muscles (GL, SOL, BF and ES). Fig. 2 illustrates typical EMG patterns averaged across 9 trials by a representative subject who performed the LR_B task with the load of 5 kg. The moment of load release is shown as time zero. Intervals of EMG integration (see Eqs. (1a,b) in Section 2) are shown with vertical dashed lines. Early EMG changes (marked as APAs in Fig. 2) varied across subjects; in particular, only about half of the subjects showed clear bursts or episodes of EMG suppression in GMED and TF.

When the subject released a load held at the body side (LR_S), there was an early increase in the activity of some of the dorsal muscles. TF and GMED commonly showed an increase in the activity, but these patterns could vary significantly across subjects. Fig. 3 shows typical EMG patterns averaged over 9 trials for a representative subject who performed the LR_S task with the load of 4 kg. Note that the EMGs were recorded in muscles of the right side of the body, and the load acted upwards on the right hand extended sideways (Fig. 1B in Section 2). APAs varied across

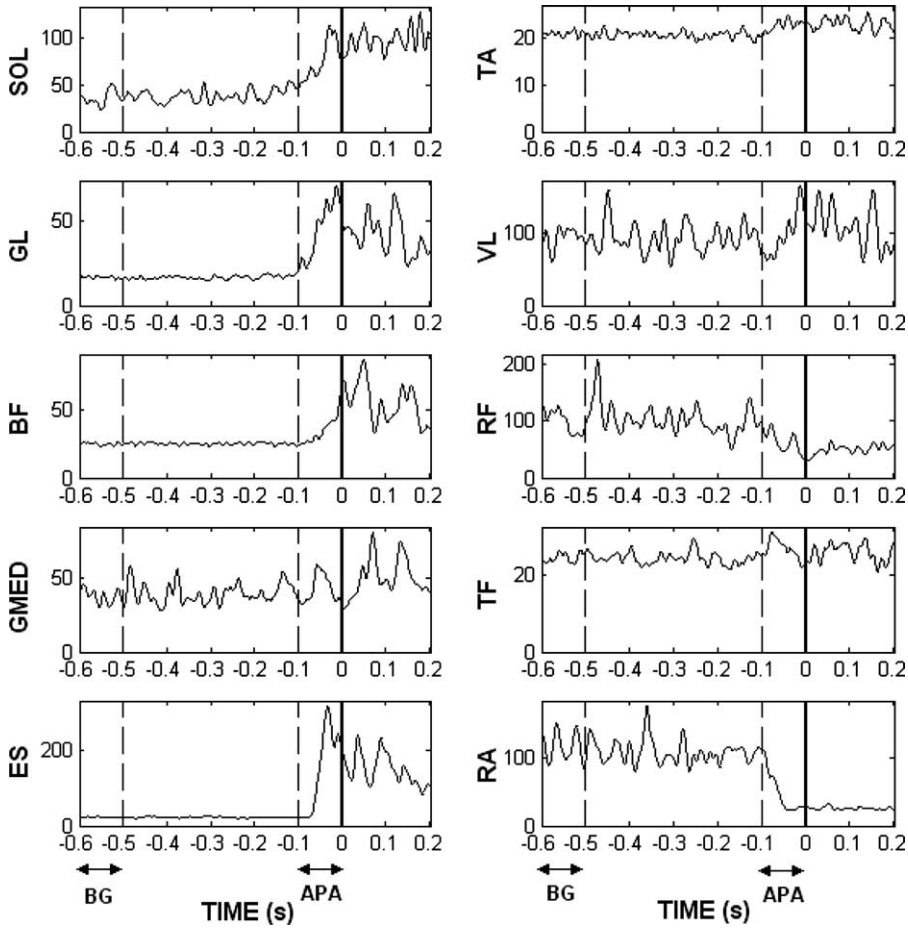

Fig. 2. EMG patterns averaged across 9 trials by a representative subject in the LR_B task. *Vertical solid lines* correspond to the alignment time, time zero, t_0. Note the early increase in the activity of dorsal muscles, accompanied sometimes by a decrease in the activity of ventral muscles. EMG scales are in arbitrary units. Muscle abbreviations are the same as in Fig. 1.

subjects, and some subjects did not show clear bursts or episodes of EMG suppression in some muscles.

3.1.2. PCA of the data obtained in the LR_B task

The indices of integrated muscle activity associated with an early shift of the COP (ΔIEMG indices, see Section 2) for all muscles were measured in each of the 63 repetitions of the LR_B task and normalized by the integrated muscle activity during control trials resulting in $\Delta IEMG_N$. The $\Delta IEMG_N$ indices for each subject were subjected to PCA. Consistent with previous studies (Krishnamoorthy et al., 2003a,b), across all subjects, we found that the first 3 PCs had at least two muscle signals significantly loaded each. Principal components from PC4 onward not only explained little variance in the $\Delta IEMG_N$ space, but these components had at most one muscle signal with significant loading, and were poorly reproducible across subjects. The first 3 PCs accounted on average for about 59.2% (±1.1%; ranging from 55.5 to 64.2%) of the total variance. The average amount of variance explained by PC1 was 27.1% (±0.8%; ranging from 24.3 to 31.7%), by PC2 was 18.0% (±1.0%; ranging from 15.6 to 23.2%) and by PC3 was 14.0% (±0.4%; ranging from 12.4 to 15.4%) across all subjects (Table 1). PCs were ranked according to the percentages of the total variance explained across subjects. Table 2 shows the loadings of all the muscles on the 3 PCs for a representative subject. The significant loadings (with the absolute value over 0.5; Hair et al., 1995) are shown in bold. Muscles that typically showed significant loadings in the 3 PCs were:

PC1: GL, SOL, BF, and ES (push-back M-mode or M_{1B}-mode)
PC2: TA, VL, RF, TF, and RA (push-forward M-mode or M_{2B}-mode)
PC3: GMED and TF (push-side M-mode or M_{3B}-mode)

The labels of the muscle groups should not be taken literally: We used apparent action of the significantly loaded muscle groups on the center of mass along the two major axes, anterior–posterior and medio-lateral, to classify the modes.

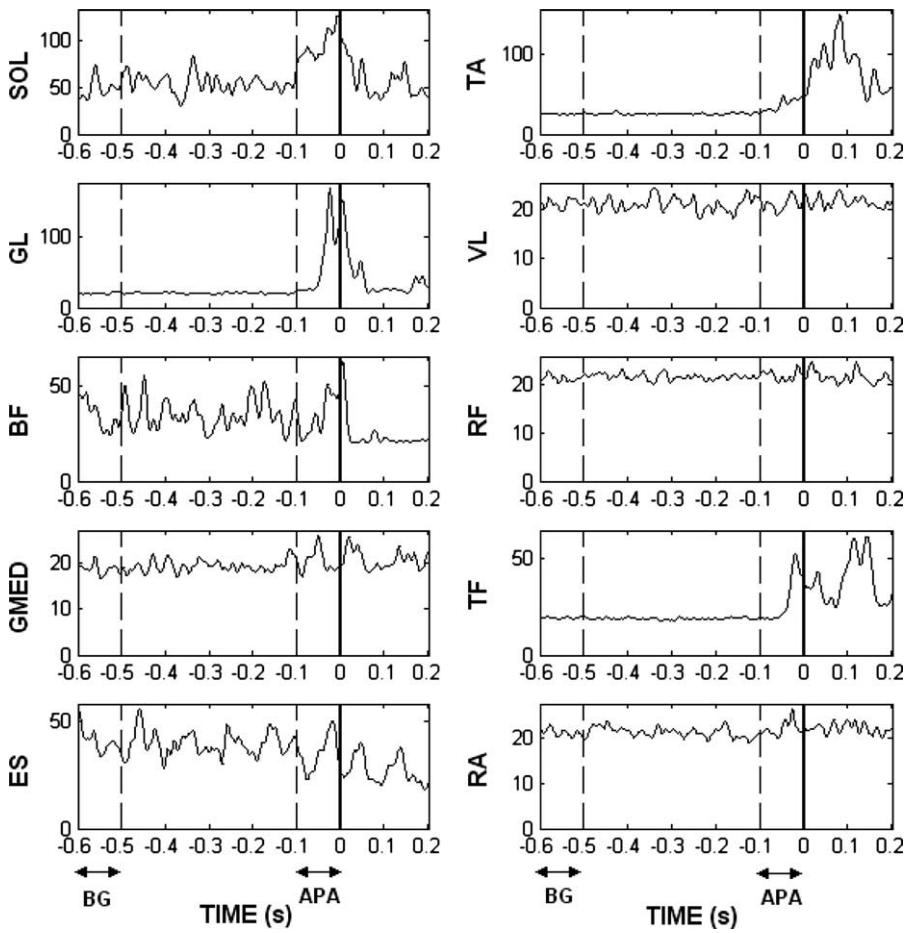

Fig. 3. EMG patterns averaged across 9 trials by a representative subject in the LR_S task. *Vertical solid lines* correspond to time zero, t_0. Note the increase in the background activity of the dorsal and lateral muscles. Abbreviations are the same as in Fig. 1.

3.1.3. PCA of the data obtained in the LR_S task

Similar computations applied to the data obtained in the LR_S task resulted in retaining 3 PCs for two subjects and 4 PCs for 6 others, which explained about 60.0% (±1.3%; ranging from 53.3 to 63.2%) of the total variance. The average amount of variance explained by PC1 was 19.7% (±0.9%; ranging from 16.5 to 25.1%), by PC2 was 16.4% (±0.4%; ranging from 14.7 to 17.7%), by PC3 was 14.6% (±0.5%; ranging from 13.0 to 16.5%), and by PC4 was 12.5% (±0.4%; ranging from 10.9 to 13.9%) across all subjects (Table 1). Table 3 shows the loadings of all the muscles on the PCs for a representative subject in the LR_S task. The significant loadings are shown in bold. Muscles typically seen significantly loaded in the PCs were:

PC1: GL, SOL, BF, and ES (push-back M-mode or M_{1S}-mode)
PC2: GMED, TF, *TA*, *VL and BF* (push-side M-mode or M_{2S}-mode)
PC3: TA, VL, RF, TF, and RA (push-forward M-mode or M_{3S}-mode)

Table 1
Results of the PCA

Muscle		PC1 (M_1-mode) ($n_{AP}=8$, $n_{ML}=8$)	PC2 (M_2-mode) ($n_{AP}=8$, $n_{ML}=8$)	PC3 (M_3-mode) ($n_{AP}=8$, $n_{ML}=8$)	PC4 (M_4-mode) ($n_{AP}=0$, $n_{ML}=6$)
AP	Mean	27.1	18.0	14.0	
	SEM	0.8	1.0	0.4	
	Range	24.3–31.7	15.6–23.2	12.4–15.4	
ML	Mean	19.7	16.4	14.6	12.5
	SEM	0.9	0.4	0.5	0.4
	Range	16.5–25.1	14.7–17.7	13.0–16.5	10.9–13.9

Mean values, standard errors of the mean (SEM), and ranges are shown for the percentages of the total variance accounted for by the first 3 or 4 principal components; *n* shows the number of subjects who showed these PCs. AP, anterior–posterior direction; ML, medio-lateral direction.

Table 2
Results of PCA during the LR_B series in a representative subject

Muscle	PC1 (M_{1B}-mode)	PC2 (M_{2B}-mode)	PC3 (M_{3B}-mode)
TA	0.1188	**0.7695**	−0.0228
GL	**0.5108**	0.0081	0.2138
SOL	**0.8468**	−0.0947	−0.0278
VL	−0.1152	**0.8546**	0.3182
RF	−0.1769	**0.8402**	0.2354
BF	**0.7875**	0.0469	−0.1876
GMED	0.4221	0.3618	**−0.5963**
TF	0.2848	0.3035	**0.6879**
RA	−0.0843	0.2119	**0.5121**
ES	**0.9093**	−0.0885	−0.1496

Loadings over 0.5 are shown in bold. TA, tibialis anterior; GL, lateral head of gastrocnemius; SOL, soleus; VL, vastus lateralis; RF, rectus femoris; BF, biceps femoris; GMED, gluteus medius; TF, tensor fasciae latae; ES, erector spinae; RA, rectus abdominis.

PC4: *SOL*, *BF*, and *ES* (second push-back M-mode or M_{4S}-mode)

The muscles indicated in *italics* in the second and fourth PC, did not show up consistently in the PC indicated, but were sometimes in one of the other M-modes.

3.1.4. Identifying the Jacobians: results of multiple regression analyses

The linear relations between changes in the magnitudes of M-modes (ΔMs) and the associated COP shifts (ΔCOP) were computed using multiple regression analysis. This was done using data from the LR_B and LR_S tasks separately. As a result, one set of regression coefficients was computed for the COP_{AP} shift and two sets of regression coefficients were computed for the COP_{ML} shift. Both M_1- and M_2-modes were significant predictors of ΔCOP magnitude. For the LR_B tasks, variations in the magnitudes of the 3 M-modes accounted for 87.0% (±3.6%) of the total variance in ΔCOP. For the LR_S tasks, variations in the magnitudes of the 3 (or 4) M-modes accounted for 74.2% (±3.8%) of the total variance in ΔCOP.

Table 3
Results of PCA during the LR_S series in a representative subject

Muscle	PC1 (M_{1S}-mode)	PC2 (M_{2S}-mode)	PC3 (M_{3S}-mode)	PC4 (M_{4S}-mode)
TA	**0.6843**	0.0720	−0.1101	−0.0120
GL	**0.7646**	−0.0728	−0.0620	0.1882
SOL	**0.8256**	−0.0686	0.0570	0.0033
VL	−0.1043	−0.0470	−0.1780	**0.8592**
RF	−0.3360	0.0166	−0.1477	**−0.5257**
BF	0.1049	−0.2443	**0.8196**	−0.0501
GMED	−0.1105	**0.8067**	−0.0181	0.0143
TF	0.1007	**0.7967**	0.0623	−0.0008
RA	−0.0094	0.2794	0.4608	**0.4659**
ES	−0.2335	0.2719	**0.5818**	−0.0018

Loadings over 0.5 are shown in bold. Muscle abbreviations are the same as in Table 1.

3.2. Analysis of the stepping (ST) tasks

3.2.1. General EMG patterns in the ST tasks

In ST trials, EMG patterns preceding a step were different in the muscles of the stepping leg and of the supporting leg. Fig. 4 shows EMG patterns in the right side of the body averaged over 20 trials at the right leg ST task performed by a typical subject. Vertical solid lines correspond to the alignment time, t_0, the time of the toe-off. EMG patterns are shown for a selected subset of muscles for the tasks of 'comfortable stepping' (ST_{CS}), 'quick stepping' (ST_{QS}) and 'reaction time stepping' (ST_{RT}). In the ST_{CS} task (left panels), the stepping leg typically showed alternating bursts of activity in TA and SOL as well as more or less in-phase changes in the activity of TF and GMED. In ST_{QS} task (middle panels), there was a substantial increase in the level of activity of most muscles, while the timing of the EMG bursts shifted closer to the toe-off time as compared to ST_{CS}. In ST_{RT} task (right panels), almost all the muscles showed substantially increased EMG bursts, which were shifted towards the toe-off time, with more pronounced modulation of EMGs in most muscles as compared to the other ST tasks. The EMG patterns were variable across the subjects but showed a general trend of later and more pronounced bursts in the sequence ST_{CS}–ST_{QS}–ST_{RT}.

Fig. 5 shows typical EMG patterns in muscles at the right side of the body during stepping with the left leg averaged across 20 trials by a representative subject. In the ST_{CS} task (left panels), alternating activation of SOL and TA was accompanied by simultaneous activation of TF and GMED. In the ST_{QS} task (middle panels), there was a substantial increase in the level of activity of the ventral muscles. EMGs in the ST_{RT} task (right panels) were mostly similar to those in the ST_{QS} task, while TF and GMED showed smaller EMG modulation as compared to the ST_{CS} and ST_{QS} tasks.

3.2.2. Changes in COP and Fx shifts

Across the 3 ST tasks, the subjects showed significantly different patterns of changes in 3 mechanical variables, COP coordinates in the AP and ML directions and F_X. Fig. 6 shows COP and F_X shift patterns averaged across 20 trials by a representative subject in the 3 ST tasks. Solid, thin, and dashed lines indicate the ST_{CS}, ST_{QS}, and ST_{RT} tasks respectively. In preparation to stepping during the ST_{CS} task, subjects shifted the COP in the ML direction first towards the foot of the stepping leg and then towards the support foot and in the AP direction backwards. These adjustments apparently allowed to unload the stepping leg and to create a moment of the reactive force rotating the body forward about the ankle joints.

Typical early $COP_{AP,}$ $COP_{ML,}$ and F_X shifts observed in the ST_{CS} task gradually became greater in the ST_{QR} and ST_{RT} tasks. COP shifts in both AP and ML directions started earlier during the ST_{CS} task as compared to the ST_{QR} and ST_{RT} tasks. The timing of the F_X shift did not differ by much

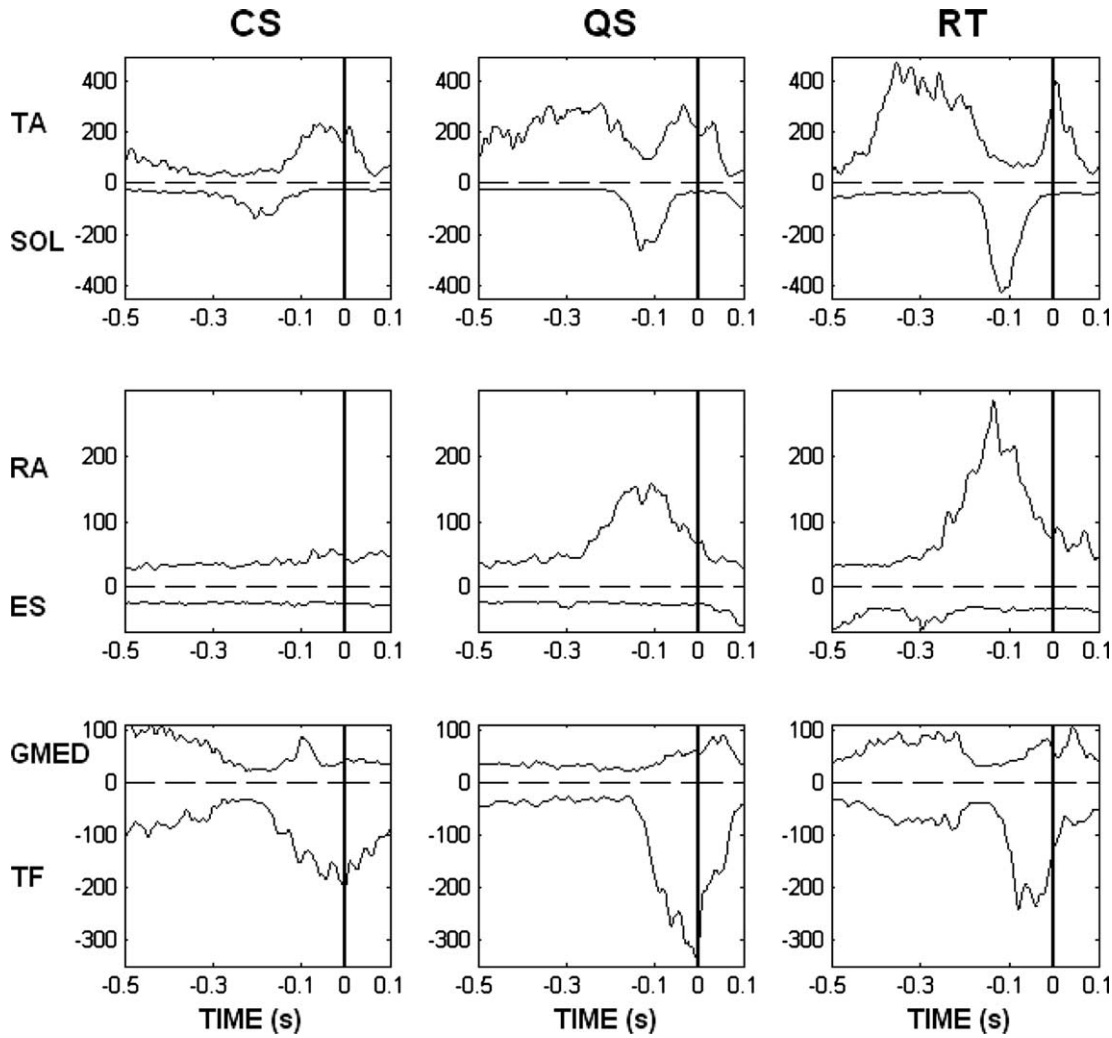

Fig. 4. EMG patterns during stepping with the right leg averaged across 20 trials by a representative subject. *Vertical solid lines* correspond to the alignment time, time zero (i_0). CS, QS and RT indicate the 'comfortable stepping' (ST_{CS}), 'quick stepping' (ST_{QS}) and 'reaction time stepping' (ST_{RT}) tasks, respectively. Note the increased muscle activity in the ST_{QS} and ST_{RT} conditions. Abbreviations are the same as in Fig. 1.

across the 3 ST tasks. The rate of change of COP_{AP} and F_X was greater in the ST_{QR} and ST_{RT} tasks as compared to the ST_{CS} task, but this was not obvious for the rate of change of COP_{ML}.

Fig. 7 shows 3 characteristics of COP and F_X shifts during the ST tasks averaged across the subjects and across the trials performed by the right leg (EMGs were recorded in the stepping leg) and by the left leg (EMGs were recorded in the support leg). The open bars correspond to COP_{AP}, the grey bars show the data for COP_{ML}, and the black bars show the data for F_X. The top panel shows the magnitude of the COP and F_X total displacement (ΔCOP and ΔF_X) within the time window from 1000 ms prior to t_0 up to t_0. The middle panel shows the time of the initiation of COP and F_X shifts defined as the time when the corresponding first derivative ($dCOP_{AP}/dt$, $dCOP_{ML}/dt$, or dF_X/dt) first exceeded 5% of its peak magnitude. The bottom panel illustrates the peak values of $dCOP_{AP}/dt$, $dCOP_{ML}/dt$, and dF_X/dt.

In the ST_{QS} and ST_{RT} tasks, the amplitude of the COP_{AP} and F_X shifts increased substantially as compared to the ST_{CS} task. In contrast, the amplitude of COP_{ML} shift in the ST_{QS} and ST_{RT} tasks was smaller as compared to the ST_{CS} task. The onset of the COP_{AP} and COP_{ML} shifts was later in the ST_{QS} and ST_{RT} tasks as compared to the ST_{CS} task, while the onset of the F_X shift showed a general trend of becoming earlier in the sequence ST_{CS}–ST_{QS}–ST_{RT}. In the ST_{QR} and ST_{RT} tasks, the peak magnitudes of $dCOP_{AP}/dt$ and dF_X/dt showed greater values than in the ST_{CS} task. However, the peak magnitude of $dCOP_{ML}/dt$ was smaller in the ST_{QR} and ST_{RT} tasks as compared to the ST_{CS} task.

One-way ANOVA with the factor *Condition* (ST_{CS}, ST_{QS}, and ST_{RT}) was used to test the significance of the differences illustrated in Fig. 7. There was a significant effect of *Condition* on ΔCOP for COP shifts in both AP direction ($F_{[2,7]}=190.0$, $P<0.001$) and ML direction ($F_{[2,7]}=3.379$, $P<0.05$). Tukey's HSD tests showed that

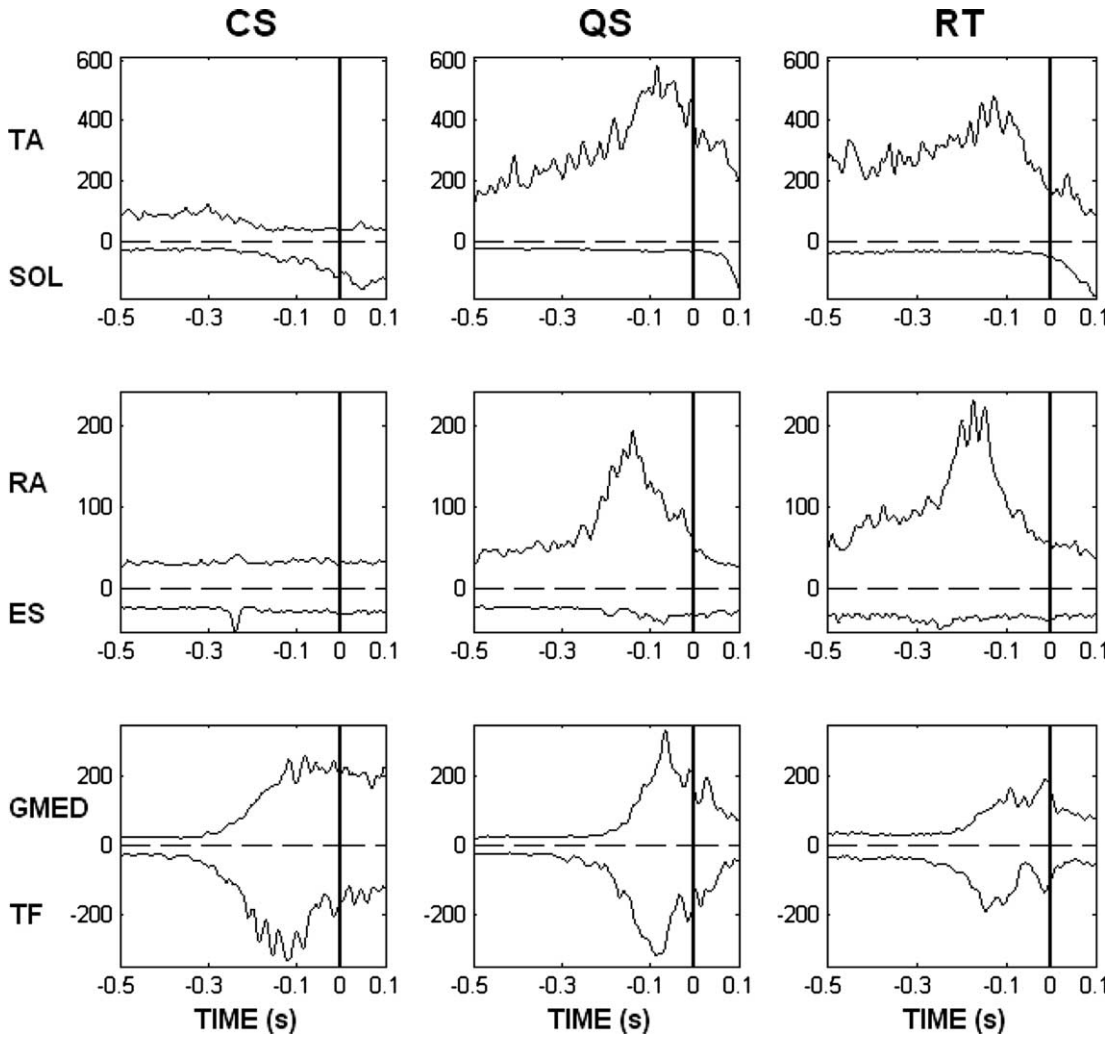

Fig. 5. EMG patterns during stepping with the left leg averaged across 20 trials by a representative subject. *Vertical solid lines* correspond to the alignment time, time zero (t_0). Note a substantial increase in the activity of the ventral muscles in ST_{QS}. Abbreviations are the same as in Figs. 1 and 4.

ΔCOP_{AP} in ST_{CS} task were significantly smaller that in both ST_{QS} and ST_{RT} tasks ($P<0.05$). There was also a significant difference in ΔCOP_{AP} between ST_{QS} and ST_{RT} tasks ($P<0.05$). Significant differences for ΔCOP_{ML} were seen between the ST_{CS} and ST_{QS} tasks ($P<0.05$) and between the ST_{QS} and ST_{RT} tasks ($P<0.05$). There was a significant effect of *Condition* on ΔF_X ($F_{[2,7]}=41.140$, $P<0.001$). Tukey's HSD tests showed that there was a significantly higher ΔF_X in the ST_{QS} task as compared to the ST_{CS} task ($P<0.05$) and in the ST_{RT} task as compared to the ST_{CS} task ($P<0.05$).

Significant effects of *Condition* were also found for the onset timing of the COP_{AP} shift ($F_{[2,7]}=8.85$, $P<0.01$), of the COP_{ML} shift ($F_{[2,7]}=49.984$, $P<0.01$), and of the F_X shift ($F_{[2,7]}=14.101$, $P<0.001$). Tukey's HSD tests showed significant differences between all 3 tasks, ST_{CS}, ST_{QS}, and ST_{RT} task ($P<0.05$) for both COP_{AP} and COP_{ML} shifts. There was a significant difference in ΔF_X between the ST_{CS} and ST_{QS} tasks ($P<0.05$) and between the ST_{CS} and ST_{RT} tasks ($P<0.05$).

One-way ANOVA on the peak rate of change of the 3 mechanical variables showed main effects of *Condition* for $dCOP_{AP}/dt$ ($F_{[2,7]}=36.85$, $P<0.01$), $dCOP_{ML}/dt$ ($F_{[2,7]}=7.71$, $P<0.05$), and dF_x/dt ($F_{[2,7]}=13.017$, $P<0.05$). Tukey's HSD tests confirmed that peak $dCOP_{AP}/dt$ increased in the sequence ST_{CS}–ST_{QS}–ST_{RT}. The peak $dCOP_{ML}/dt$ in the ST_{CS} task was significantly greater than in the ST_{QS} task ($P<0.05$), and it was higher in the ST_{RT} task that in the ST_{QS} task ($P<0.05$). There was a significant increase in dF_X/dt between ST_{CS} and ST_{QS} tasks ($P<0.05$) and between ST_{CS} and ST_{RT} tasks ($P<0.05$).

3.2.3. UCM analysis

Data from the ST tasks were used to perform analysis of the structure of variability across trials in the space of M-modes. Analyses for the two directions of the COP shift were performed separately. The total variance in the M-mode space across trials was partitioned into two components. The first component (V_{UCM}) was within an uncontrolled manifold (UCM) computed as the null-space

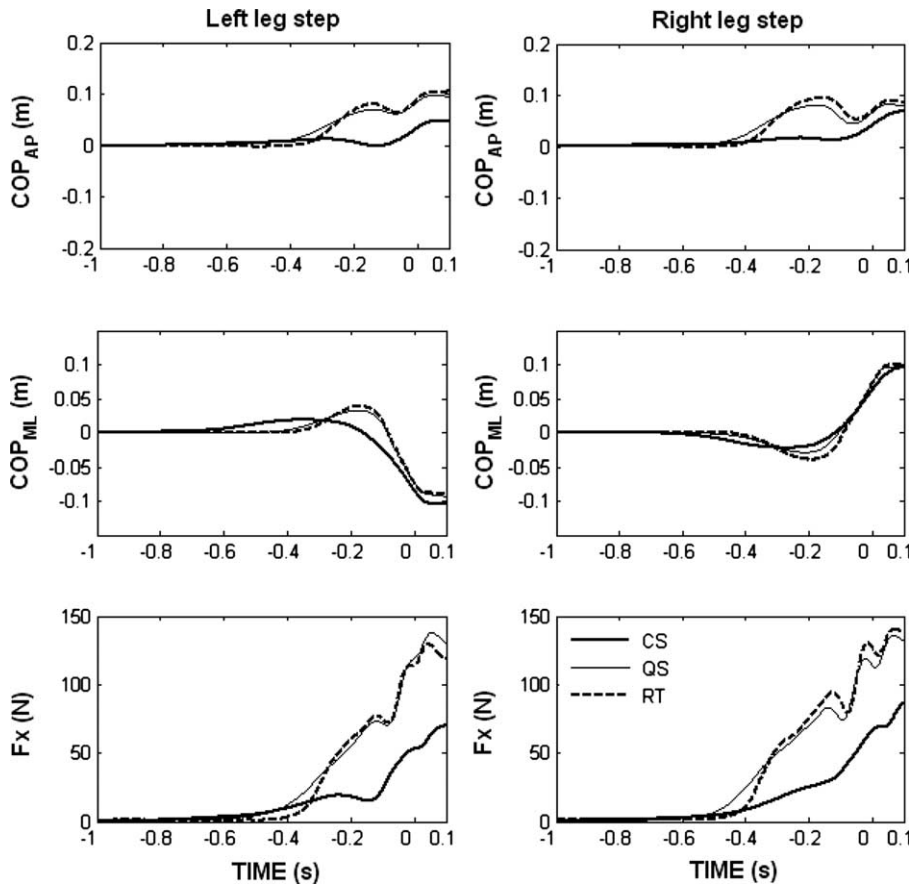

Fig. 6. COP and F_x patterns averaged across 20 trials by a representative subject in the ST tasks. Solid, thin, and dashed lines indicate the ST_{CS}, ST_{QS}, and ST_{RT} tasks respectively. Typical early COP and F_x shifts in the anterior–posterior (AP) direction observed in the ST_{CS} condition gradually became greater for the ST_{QR} and ST_{RT} tasks. However, early COP shifts in the medio-lateral (ML) direction observed in the ST_{CS} condition gradually became smaller for the ST_{QR} and ST_{RT} tasks.

of a corresponding **J** matrix defined at the previous step. The other component (V_{ORT}) was within a sub-space orthogonal to the UCM. Both V_{UCM} and V_{ORT} were quantified per degree-of-freedom in corresponding subspaces. Further, to quantify the relative amount of the total variance that is compatible with stabilization of a particular shift of the COP we used an index ΔV reflecting the difference between the variance within the UCM and orthogonal to the UCM. Positive values of ΔV showed that more variance per degree-of-freedom was within the UCM than in its orthogonal complement; we interpret this result as the existence of a multi-M-mode synergy in the M-mode space that stabilizes the average COP shift.

In general, subjects demonstrated multi-M-mode synergies stabilizing COP shifts in both AP and ML direction ($\Delta V > 0$). The ΔV index was larger for the COP_{AP} shifts and showed reproducible time changes for all 3 ST tasks. In contrast to major differences in the time profiles of mechanical variables (Figs. 6 and 7), there were relatively minor differences in ΔV across the 3 ST tasks.

Fig. 8A shows ΔV indices averaged across subjects computed for COP_{AP} shifts during the ST tasks performed by the right leg (EMGs were recorded in the stepping leg, top panels) and by the left leg (EMGs were recorded in the support leg, bottom panels). The open bars correspond to the ST_{CS} task, the grey bars show the data for the ST_{QS} task and the black bars show the data for the ST_{RT} task. Positive ΔV values indicate that most variance within a given time window was within the UCM, i.e. that an average value of COP displacement observed within that window was stabilized by covariation of magnitudes of the M-modes.

When the data were analyzed for the stepping leg (the upper panel of Fig. 8A), there were no major differences in the ΔV indices between the ST_{CS} and ST_{RT} tasks, while ΔV for the ST_{QS} task was on average higher than for the other two tasks. About 200 ms prior to the time of take-off (time zero), there was a drop in ΔV seen across all tasks. When the data were analyzed for the supporting leg (the lower panel of Fig. 8A), ΔV did not show a drop prior to the take-off time. There was a delayed increase in ΔV for the ST_{RT} task. These observations were confirmed by main effects of *Condition* (for the right leg stepping only) and *Time interval* in the Friedman test ($P < 0.05$).

Fig. 8B is organized similarly to Fig. 8A, but it illustrates ΔV indices for COP_{ML} shifts. During stepping performed with the right leg (EMG analysis in the stepping leg, the

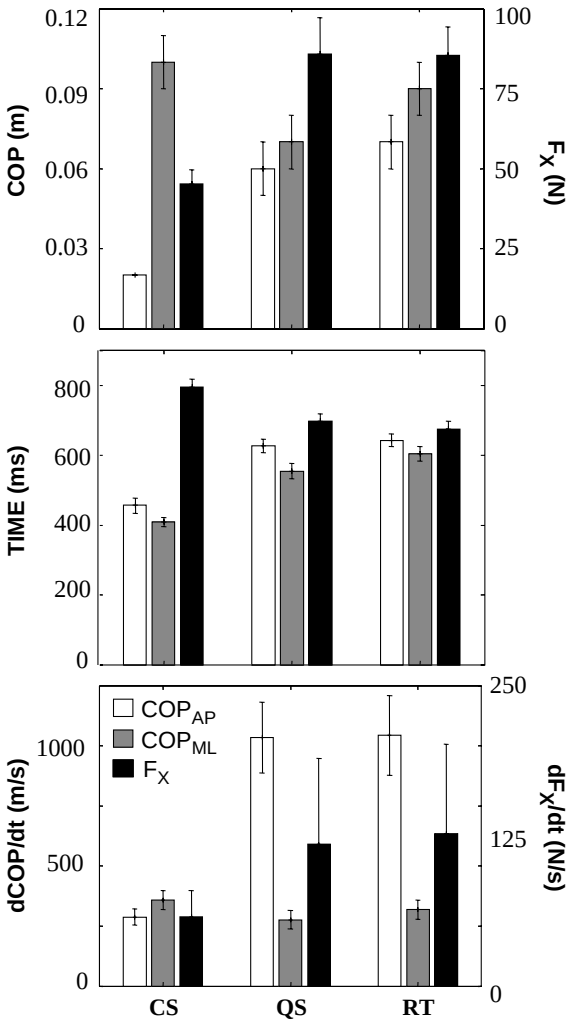

Fig. 7. Three characteristics of COP and F_X shifts during the ST tasks averaged across the subjects and across the trials performed by the right leg (EMGs were recorded in the stepping leg) and by the left leg (EMGs were recorded in the support leg). The open bars show the data for COP_{AP}, the grey bars show the data for COP_{ML}, and the black bars show the data for F_X. Standard error bars are shown.

upper panel), the general patterns of ΔV changes were similar across the 3 tasks. During stepping with the left leg (EMG analysis in the supporting leg, the lower panel), no differences across the 3 tasks were found. Both panels show a decrease of ΔV to zero or even negative values about 0.5 s prior to the take-off time across all conditions. These observations were confirmed by main effects of *Time interval* in the Friedman test ($P<0.05$).

4. Discussion

Our experiments have shown strong multi-muscle (more exactly, multi-M-mode) synergies stabilizing trajectories of the COP in both anterior–posterior (AP) and medio-lateral (ML) directions across all 3 conditions. Despite the significant differences in mechanical and EMG patterns of postural adjustments, the synergies showed only minor differences across the conditions. This result demonstrates their robustness across different speeds of making a step and manners of step initiation, self-paced and reaction time. This outcome is at least partly unexpected because of the documented major changes in the timing of anticipatory postural adjustments (APAs) under the simple reaction time conditions (Boisset and Zattara 1990; De Wolf et al., 1998; Lee et al., 1987).

4.1. Making a quick step vs. making a comfortable step

Making a fast, self-paced step was associated with postural adjustments that showed relatively minor timing differences as compared with making a comfortable self-paced step. In contrast, the magnitude of the adjustments and the rate of their changes increased significantly (Figs. 4–7). These results confirm earlier reports on relatively invariant timing of postural adjustments prior to making steps of different speed (Couillandre et al., 2002; Ito et al., 2003). In addition, we also observed significant differences in changes of COP shifts in the AP and ML directions with stepping speed. COP_{AP} showed much larger modulation with speeding-up the step than COP_{ML} (Fig. 7). This makes mechanical sense. The overall magnitude of COP shift in the ML direction should not depend on the speed of stepping because in all conditions it should ultimately lead to shifting the weight of the body onto the supporting leg. Since the timing of the postural adjustments shows minimal changes with the speed of stepping, the rate of COP_{ML} shift is also expected to remain unchanged. In contrast, COP_{AP} shifts (and also changes in the horizontal force in the AP direction, F_X, see Fig. 7) are associated with motion of the body forward and are expected to be larger and faster to allow the body to perform a faster step.

Despite the significant changes in EMG patterns and COP_{AP} shifts with stepping speed, analysis of multi-muscle synergies stabilizing these shifts has shown minimal differences. For both comfortable and quick stepping, synergies stabilized COP displacement in the AP direction at least 1 s prior to the take-off. Indices of this stabilization (ΔV) were similar across the two conditions. In both conditions, the index of COP_{AP} stabilization showed a decline in the muscles of the stepping leg about 200 ms prior to the take-off. As argued in an earlier paper (Wang et al., in press), this strategy is sensible since it phases-out the role of the stepping leg muscles in COP stabilization in preparation to loss of contact between this leg and the ground.

4.2. Making a self-paced step vs. making a reaction-time step

Several studies investigated postural preparation to stepping in simple reactions time conditions. Several studies used platform motion as the stimulus to trigger a protective step (Burleigh and Horak, 1996; McIlroy and Maki, 1999).

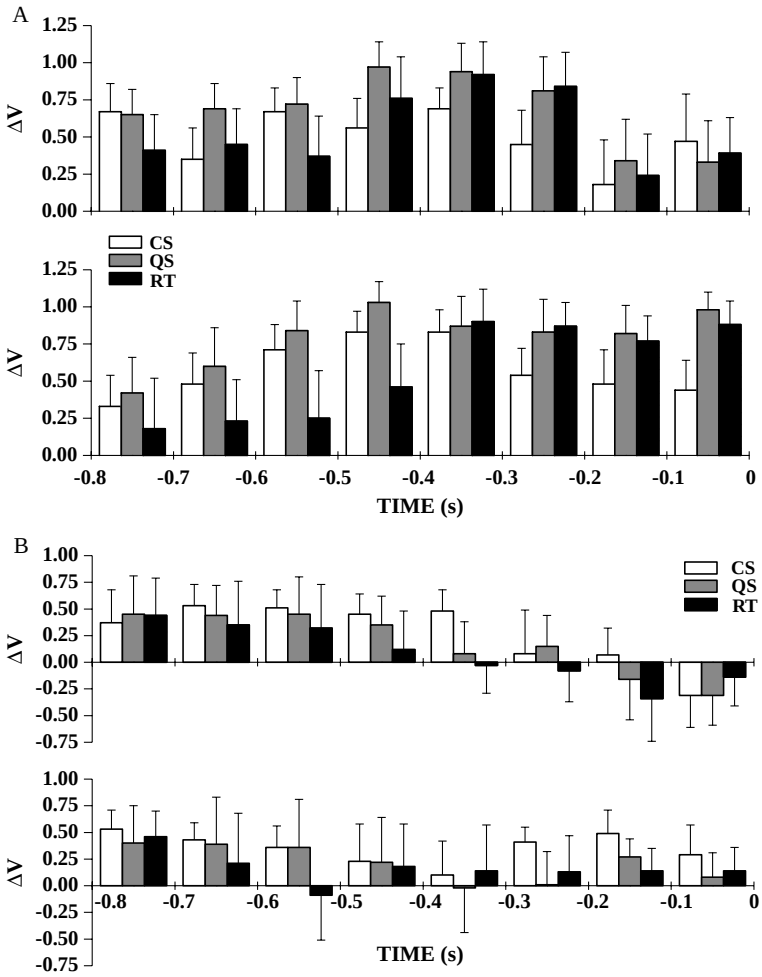

Fig. 8. (A) ΔV indices averaged across subjects computed for COP_{AP} shifts during the ST tasks performed by the right leg (EMGs were recorded in the stepping leg, top panel) and by the left leg (EMGs were recorded in the support leg, bottom panel). (B) ΔV indices averaged across subjects computed for COP_{ML} shifts during the ST tasks performed by the right leg (EMGs were recorded in the stepping leg, top panel) and by the left leg (EMGs were recorded in the support leg, bottom panel). The open bars correspond to the ST_{CS} task, the grey bars show the data for the ST_{QS} task and the black bars show the data for the ST_{RT} task. Standard error bars are shown.

One should, however, exert caution in making comparisons between those studies and experiments with voluntary step initiation in response to an imperative stimulus that by itself did not endanger posture. In particular, Burleigh and Horak (1996) have concluded that the central nervous system utilizes feed-forward prediction to modify peripherally triggered postural responses, and utilizes immediate afferent information to modify the centrally initiated postural adjustments associated with voluntary movement. Such effects were apparently impossible in our experiment when no peripheral stimulus was used to trigger postural responses.

Rogers et al. (2001) used 3 types of imperative stimuli to trigger a step. Their observations are similar to the general patterns observed in our study. In particular, they report lift-off after about 600 ms following a visual imperative stimulus in young subjects, which is close to our results. Their general patterns of COP shifts were also similar to those observed in our subjects.

Similarly to the earlier comparison between comfortable and quick steps, there is a contrast between major differences in mechanical and EMG patterns prior to stepping under the self-paced and reaction-time instructions and relatively minor differences in the patterns and magnitudes of the index of COP stabilization (ΔV, Figs. 7 and 8). The only difference is the somewhat delayed increase in the ΔV index of the M-mode synergy stabilizing COP shifts in the AP direction (Fig. 8). This result shows that the instruction was indeed effective in delaying some of the phenomena associated with step initiation. However, the relative paucity of differences across the 3 conditions in the ΔV indices suggests that the mutli-muscle synergies are robust across both speeds and method of step initiation.

4.3. Multi-muscle synergies: is this a useful concept?

The concept of postural synergies has been used in motor control for a long time (Bernstein, 1947). However, only

recently, computational methods have started to be developed to allow quantitative assessments of such synergies. Various matrix factorization methods have been applied to define synergies in the spaces of muscle activation levels (EMGs) and of kinematic variables (Saltiel et al., 2001; Ting and Mcpherson 2005; Tresch et al., 2004). In particular, principal component analysis has provided a powerful tool for identifying synergies for a variety of motor behaviors (Alexandrov et al., 1998; Ivanenko et al., 2004; Sanger, 2000; Santello et al., 1998; Shim et al., 2005). We would like to note, however, that all these studies focused on a particular aspect of synergies, namely patterns of sharing a common, functionally important performance variable among elements such as muscles, joints, or digits. In our parlance, these studies focused on identifying stable modes, i.e. groupings of apparent elements of the system. Within our framework, synergies are created in the mode space, i.e. in the space of latent, immediately not observable variables—not in the space of apparent elements—and their major feature is stabilization of an important performance variable across trials or in time during performance of a task. We would like to note here that our particular labeling of M-modes should not be taken too literally; it has been based on the apparent combined action of the significantly loaded muscles on the center of mass of the body. The M-modes have been classified as acting along two main axes, AP and ML. This is certainly a simplification made for convenience of data description. M-mode action may be diagonal; M-modes may also be associated with muscle co-contraction (as shown by Krishnamoorthy et al., 2004) with a minimal net force/torque effect.

Bernstein (1967) used the notion of synergies as the means of solving the problem of motor redundancy, i.e. of decreasing the number of variables the controller needs to manipulate. In addition, synergies may play another important role, that of modulating the degree of stabilization of performance variables. If a person wants to maintain a steady-state value of a performance variable or to change it relatively slowly, a synergy stabilizing that variable may be useful. If a person wants to change a performance variable quickly, the same synergy would resist such a change. Recently, a phenomenon of anticipatory changes in synergies has been described (anticipatory covariation, Olafsdottir et al., 2005; Shim et al., 2005) in preparation to a quick change in a performance variable. This finding emphasizes the importance of the latter role of synergies, i.e. their effects on stability (flexibility) of motor behavior. It is possible that the drop in ΔV for COP_{ML} shifts observed in our experiments close to the moment of lift-off (Fig. 8) was related to the planned quick changes in COP_{ML} during the step and touch-down and represented in fact an example of anticipatory covariation. Currently, we do not have a solid explanation for the difference in the behavior of the ΔV indices for COP_{AP} and COP_{ML} at the times close to the lift-off.

One can only speculate on the possible role of proprioceptive feedback in the M-mode synergies. On the one hand, similar earlier studies of M-mode synergies during APAs have suggested the predominant importance of feed-forward mechanisms (Krishnamoorthy et al., 2003b, 2004). However, in the current experiment, analysis was performed over time intervals that allowed contribution of proprioceptive feedback in the identified M-mode synergies.

4.4. Limitations and further directions of the study

To summarize, we view the notion of muscle synergies in its modern, more exact incarnation as an emerging powerful tool for quantitative analysis of muscle interaction over a variety of natural motor actions. This notion allows to study phenomena that have been either unknown or beyond the reach of available methods until recently.

However, there are shortcomings in the current experimental design that suggest further developments of this approach. In particular, our setup had an inherent limitation of the number of EMG channels we could record simultaneously. We have decided to record muscles only on one side of the body assuming a symmetrical pattern on the other side. This assumption may well be wrong, particularly during lateral COP shifts. We will try to overcome it in future studies. At the same time, the statistically significant results indirectly support our approach. Few muscles of the leg/trunk accessible for surface EMG have strong lateral action; we recorded two, GMED and TF. Note, however, that muscles with pronounced anterior–posterior action also typically have a lateral component of action (reviewed in Nichols, 2002). Our sets of M-modes for the medio-lateral COP shifts indeed accounted for less variance, and the results of the UCM analysis were less conclusive.

PCA defines orthogonal M-modes, which is preferred for computational reasons because such M-mode sets correspond to equal amounts of variance in all directions in the M-mode space in the absence of an organizing (synergetic) action. Using non-orthogonal sets of variables (e.g. Saltiel et al., 2001; Tresch et al., 2004) may be able to account for more variance in the original data at the first step of analysis (M-mode identification). However, it complicates interpretation of the second step (the UCM analysis) since differences in the amounts of variance in different subspaces may be due to the non-orthogonal axes rather than to a control strategy.

The current method is cumbersome and time consuming. In particular, it requires collecting numerous trials as a preparatory step to be able to identify M-modes (cf. Krishnamoorthy et al., 2003a,b). This is a major stumbling block on the route to apply the method to actions by special populations involving patients with neurological disorders and elderly. In particular, it would be important to know whether the emergence of stable synergies stabilizing COP

shifts is a prerequisite for the emergence of typical postural adjustments to step initiation. It is also not known whether changes in postural preparation to stepping are associated with impaired multi-muscle synergies as it may be expected from recent studies showing impaired multi-digit synergies in elderly during pressing and prehension tasks (Shim et al., 2004; Shinohara et al., 2004). Hence, our next challenge is to overcome this delimitation and develop a method of assessing M-modes based on only a handful of trials.

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